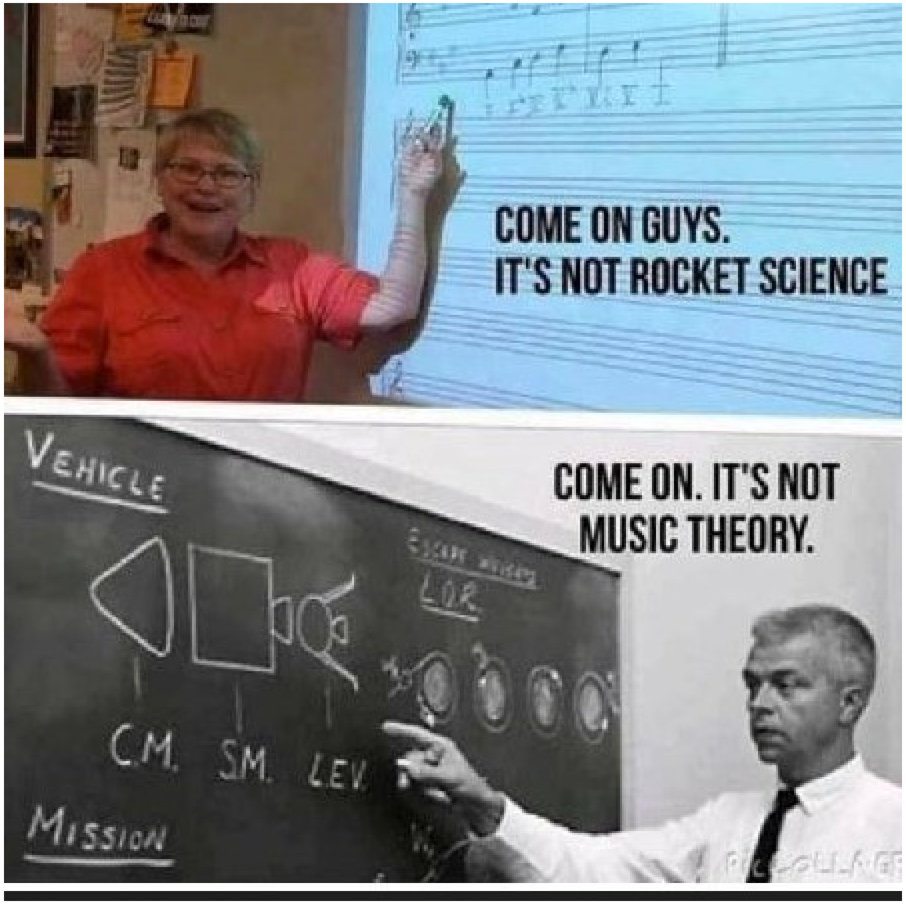
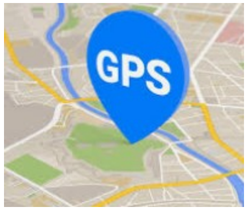
# Why you need to take Physics → Motivational speech!

Physics lays down the foundations of technology

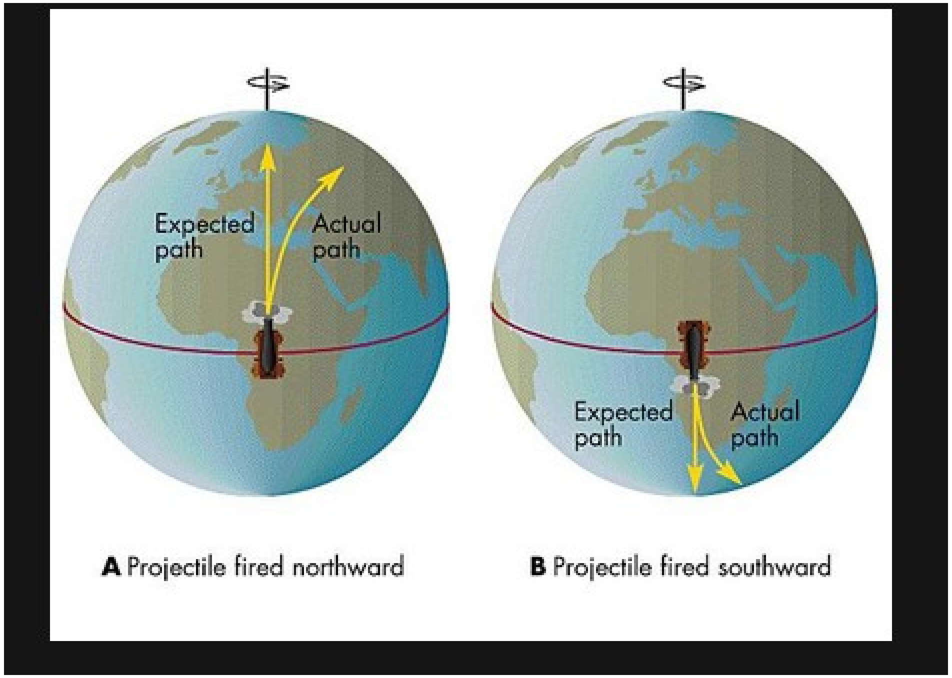
Frame of Reference!



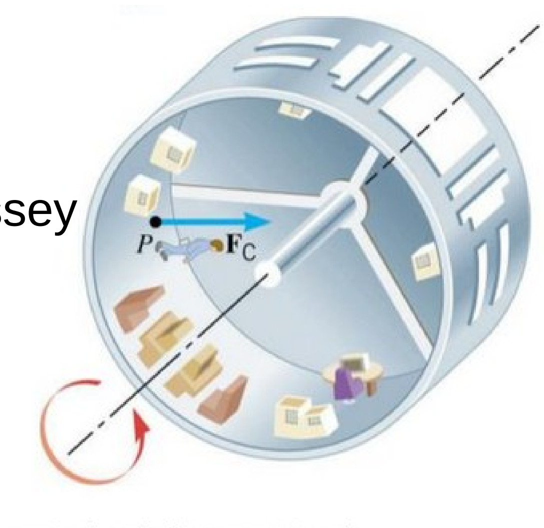

Little Boy and Fat Man | P...

Rotating frames of reference → Coriolis force and centrifugal force
(also called pseudo forces. Happen in a non inertial frame of reference)

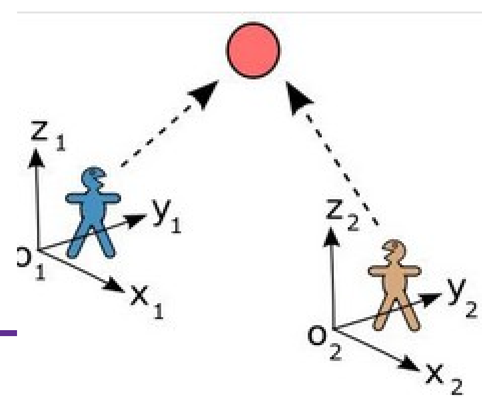



**centripetal force or centrifugal force?**
**A matter of frame of reference.**

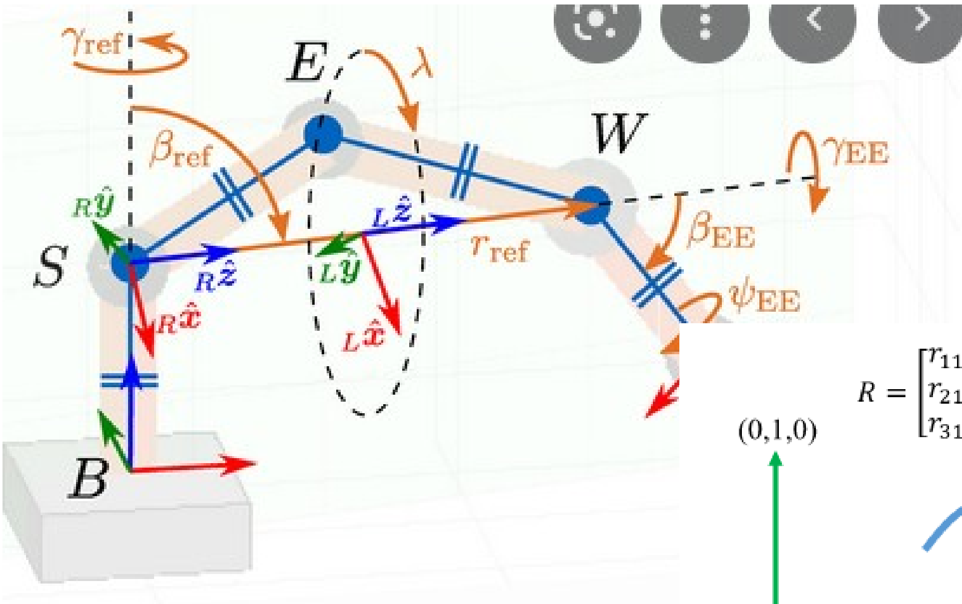
See
Space odyssey
movie

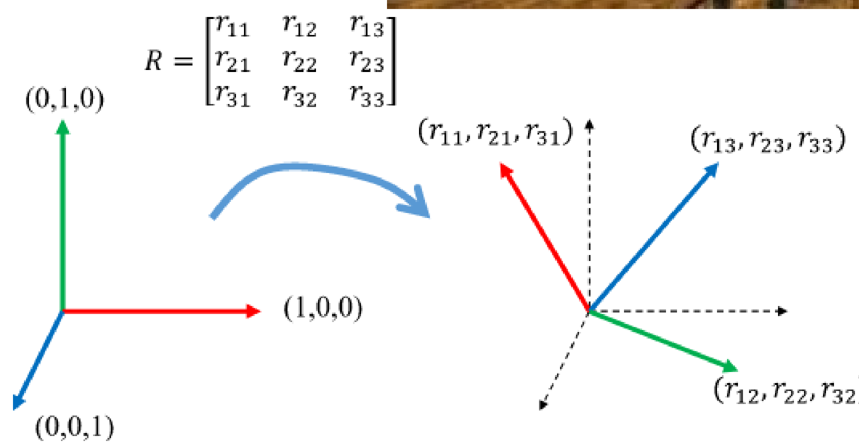



See video. In Physics, the frame of reference matters!

# Frames of reference in coding and robotics.

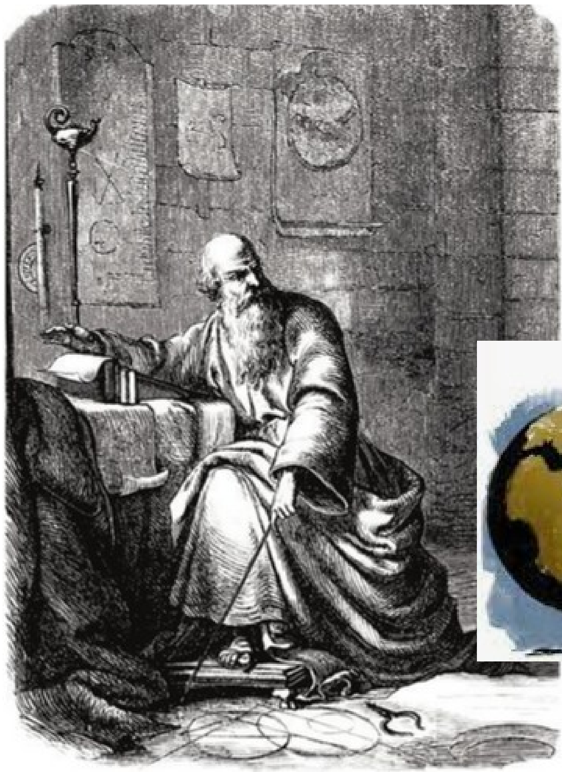


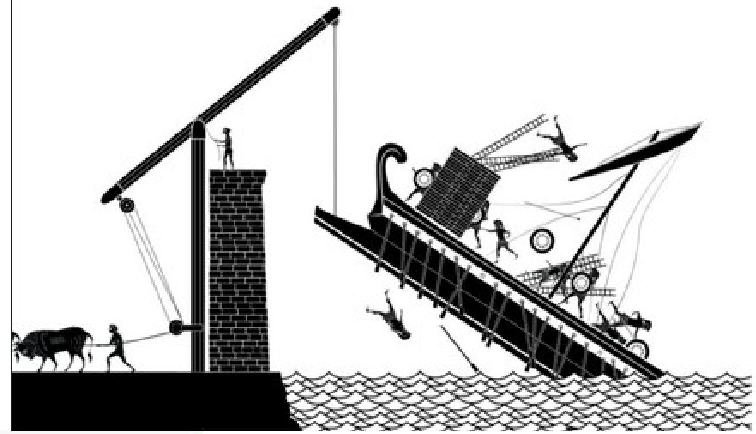




Galileo (1564-1642) the laws of nature
are written in the language of mathematics.
This was also noted thousands
of years earlier by Greeks,
Babylonians and Indians

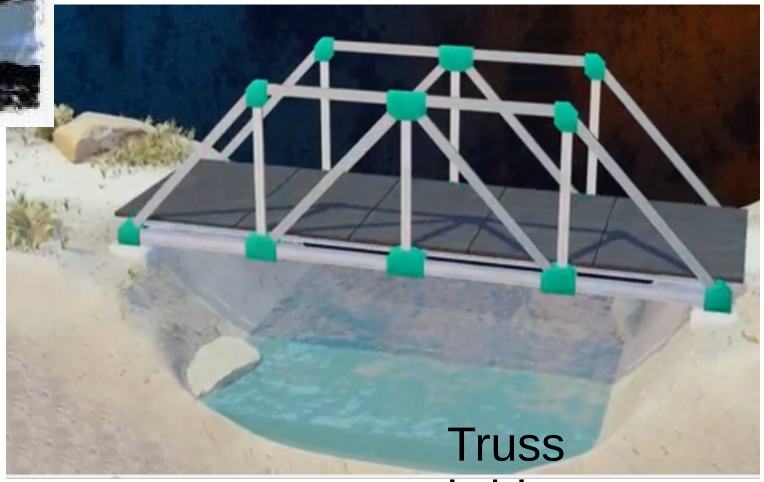
**Bing a TI 83 (TI 30) in class!**
**Review Sci Notation, proportions**
**And trig.**

# Mechanics – Equilibrium ( static ) – Machines (conservation of energy) And torques





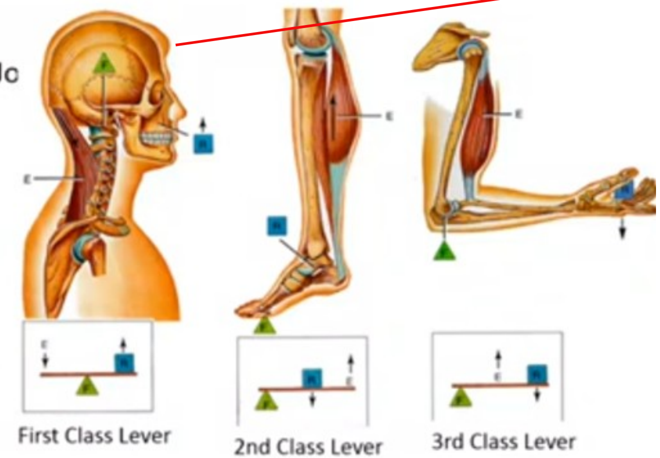
Give me a lever long enough
and a fulcrum on which to place it,
and I shall move the world.

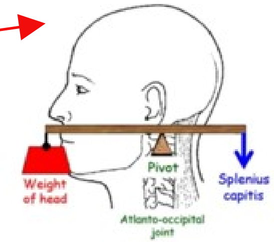


Truss bridge

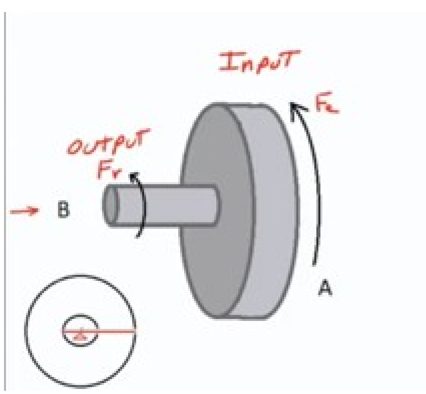
# Type of Levers in Human Body


- First Class Lever
  - Example: Head-Neck Jo
- Second Class Lever
  - Example: Ankle Joint
- Third Class Lever
  - Example: Elbow Joint

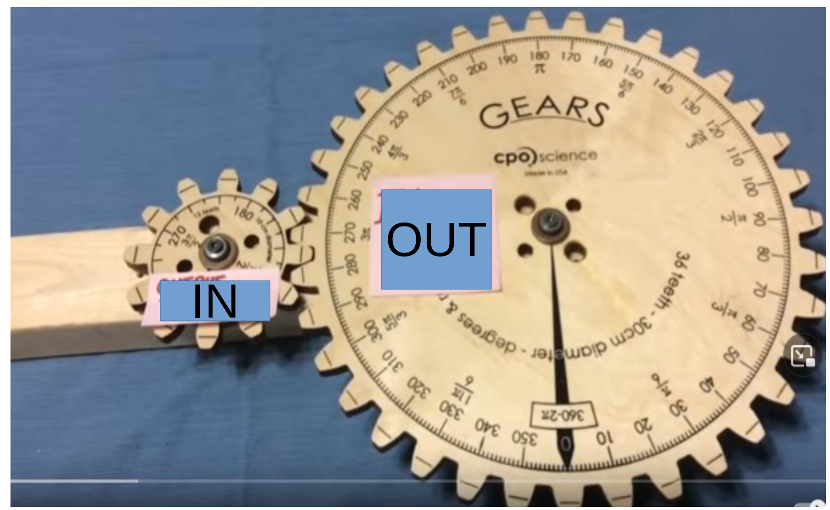



Wheel and axle

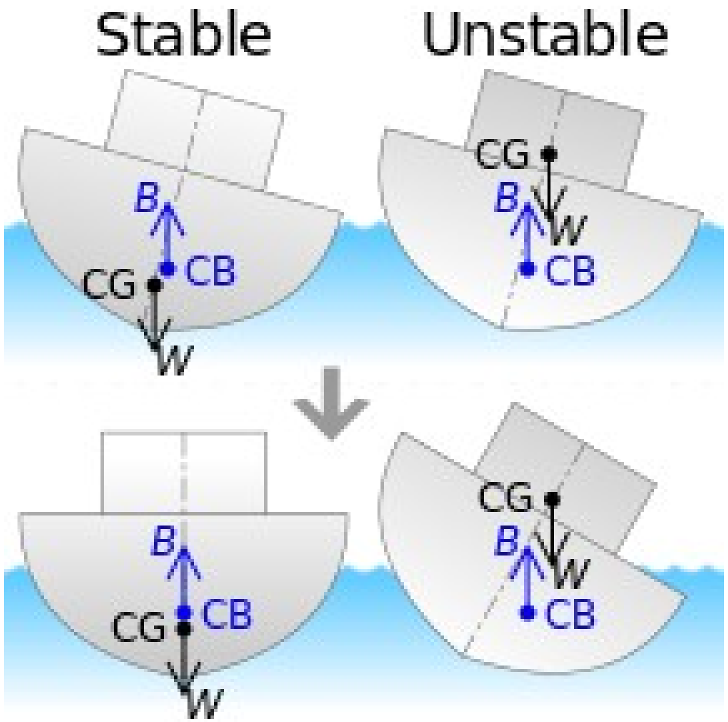




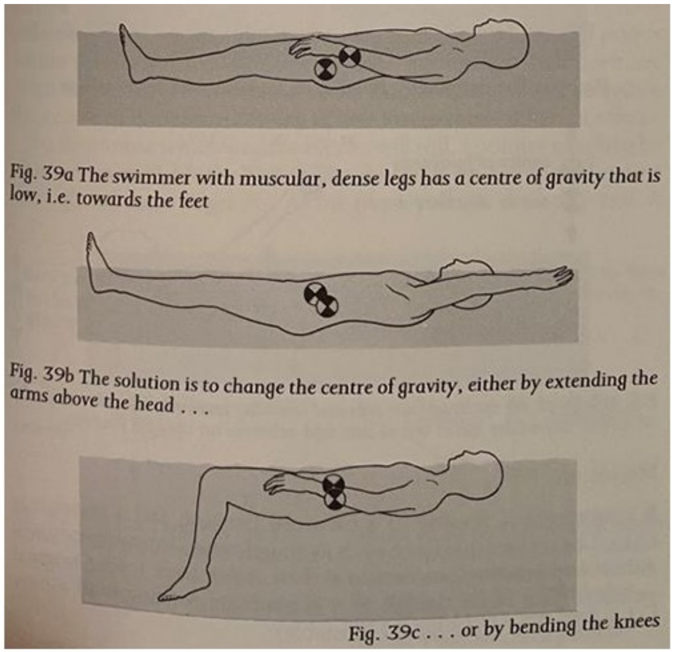



Low gear → little effort, large angle → large load , small angle

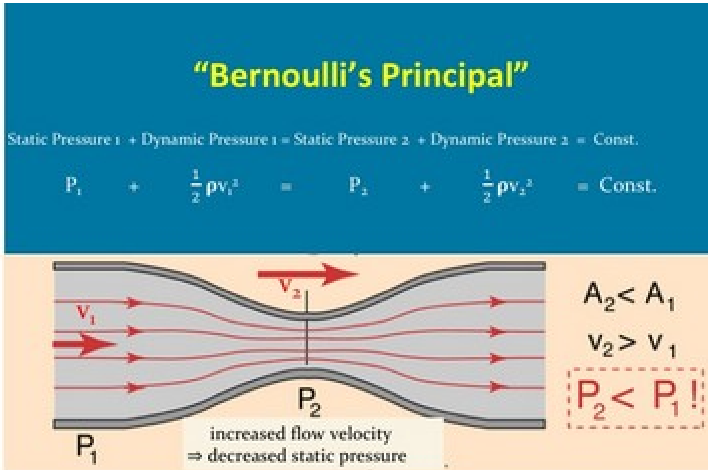
**Figure 0.4** Archimedes is best known for observing that water was displaced when he stepped into his bath. He proceeded to conduct experiments where he measured the amount of water that overflowed when objects were placed into a tub full of water. He established a principle that states that an object immersed in a fluid will experience a buoyant force equal to the weight of the displaced fluid. Archimedes is also recognized for his work with simple machines like the screw, lever, and pulley. Legend has it that Archimedes said, "Give me a firm spot on which to stand and I will move the earth" (referring to the use of a lever).



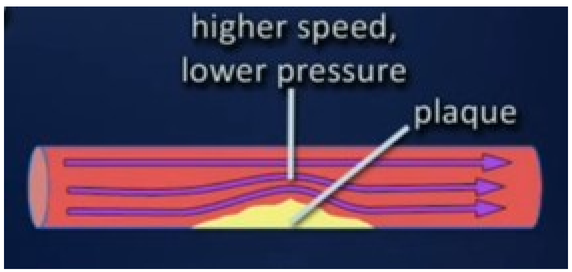

## Fluid mechanics



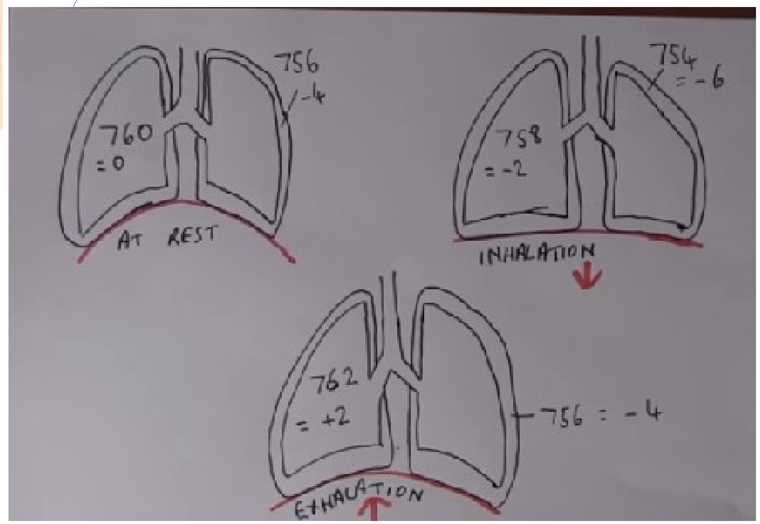

*Applied Physics*, 11e
Dale Ewen, Neill Schurter, and P. Erik Gundersen



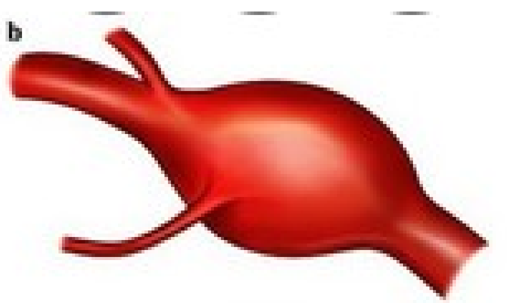




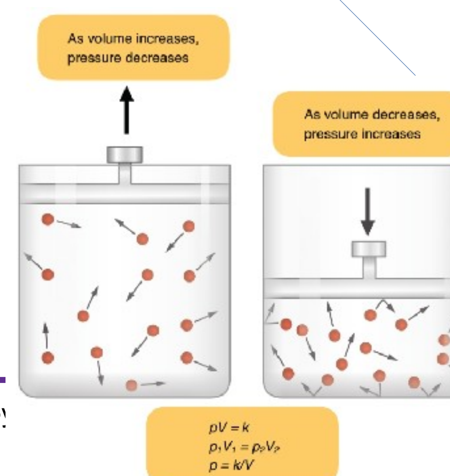

Arterial stenosis:

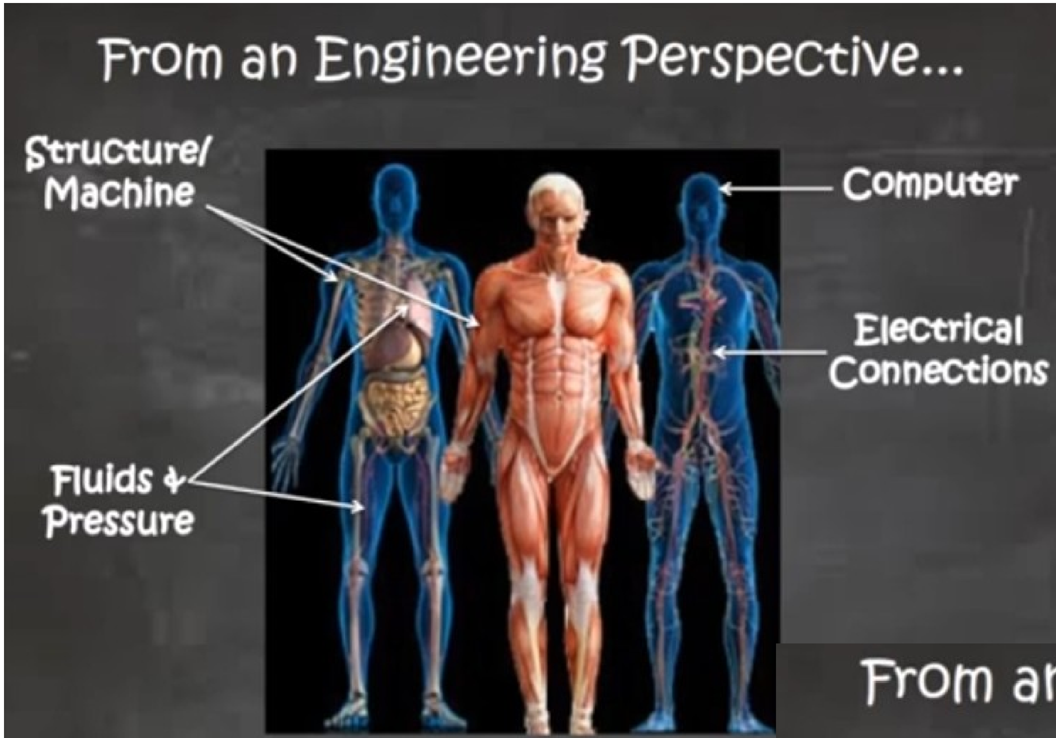



Thermodynamics
Boyle's
law




Fusiform Aneurysm



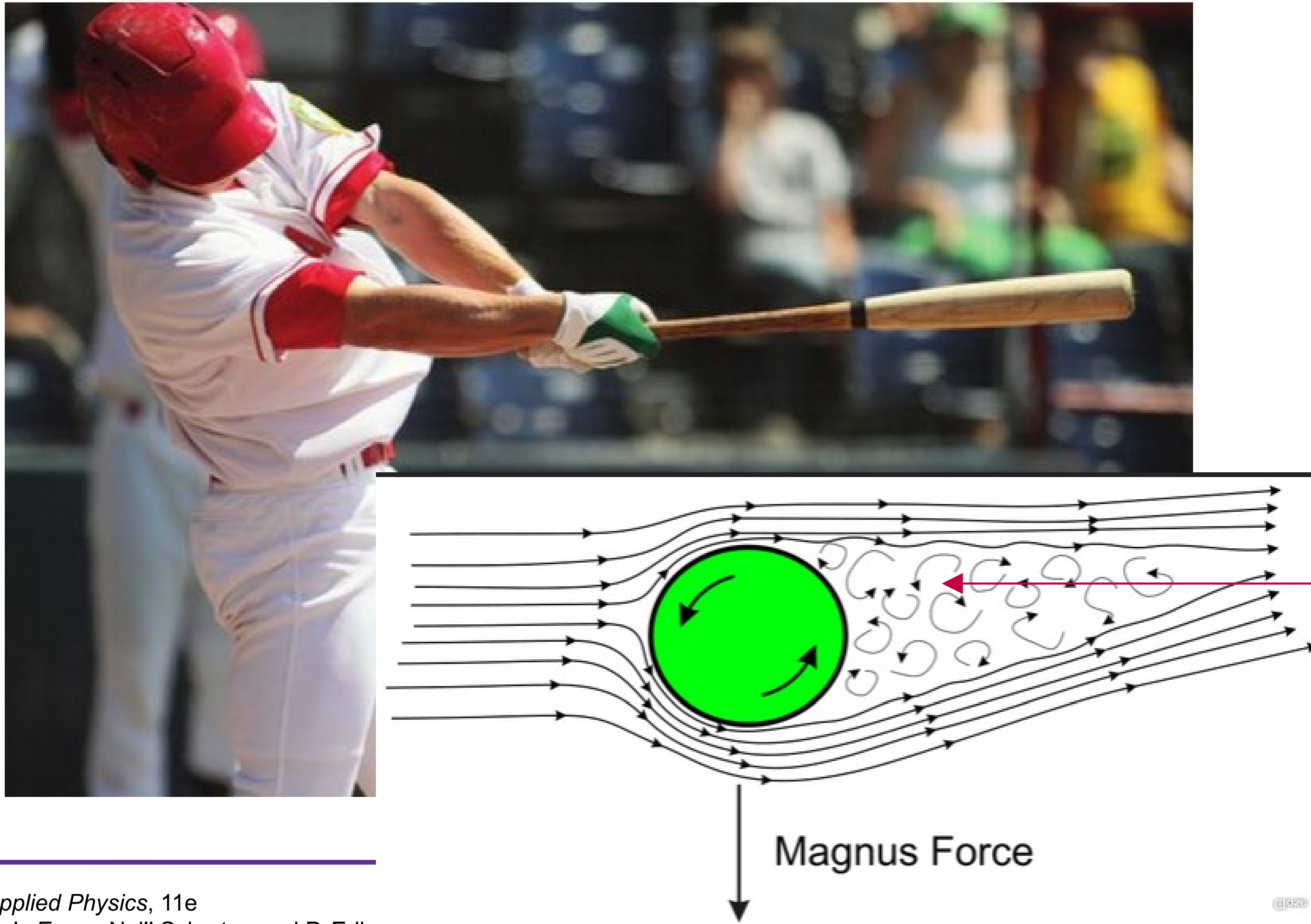



## From an Engineering Perspective...

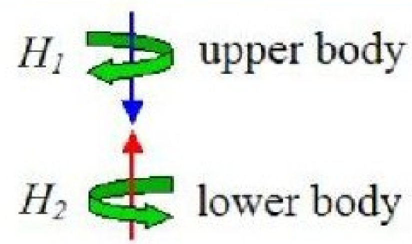
- The musculoskeletal system is a structure and machine that provides balance and motion.
- Functions of the cardiovascular system (breathing and pulse) are governed by laws of fluids and pressure.
- The brain is a computer with an electrical connection called the nervous system.

**Figure 0.2** A baseball player's understanding of physics can help improve all aspects of his game, including pitching, batting, and fielding. Photo courtesy of fotolia © Paul Yates. Reprinted with permission

Video



Angular momentum, torque, conservation of momentum,


$$H_1 + H_2 = 0$$

**Figure 0.14** Each ski design is created for use based on scientific knowledge. Photo courtesy of fotolia © Lulu Berlu

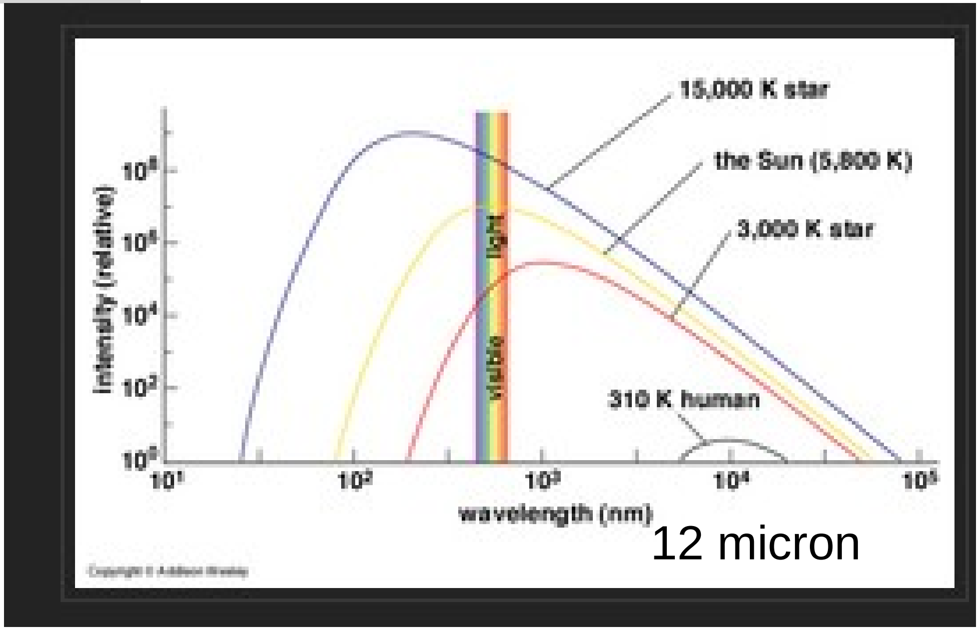
Friction
Air resistance

# Heat


Black body

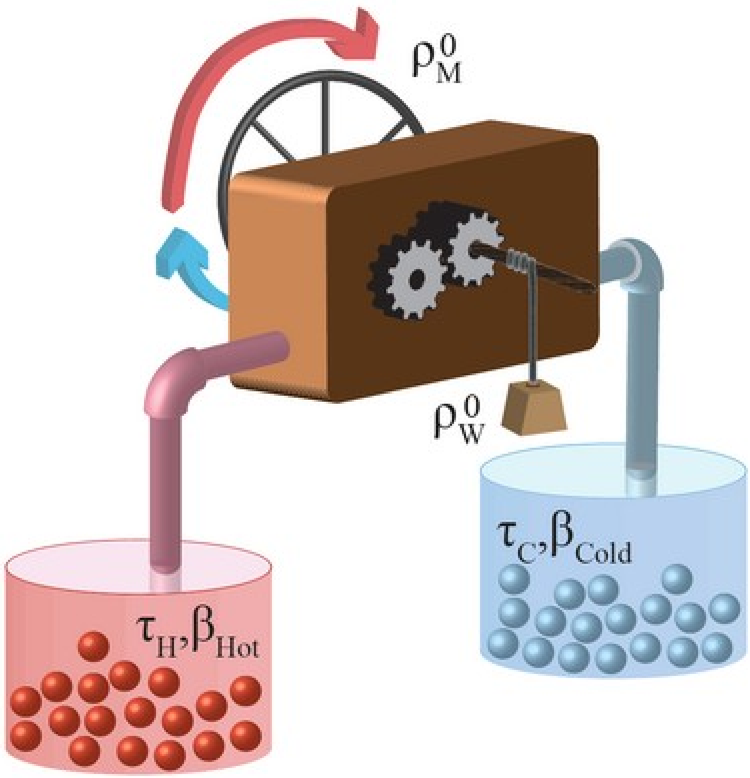
We are on average like
A 100W light bulb

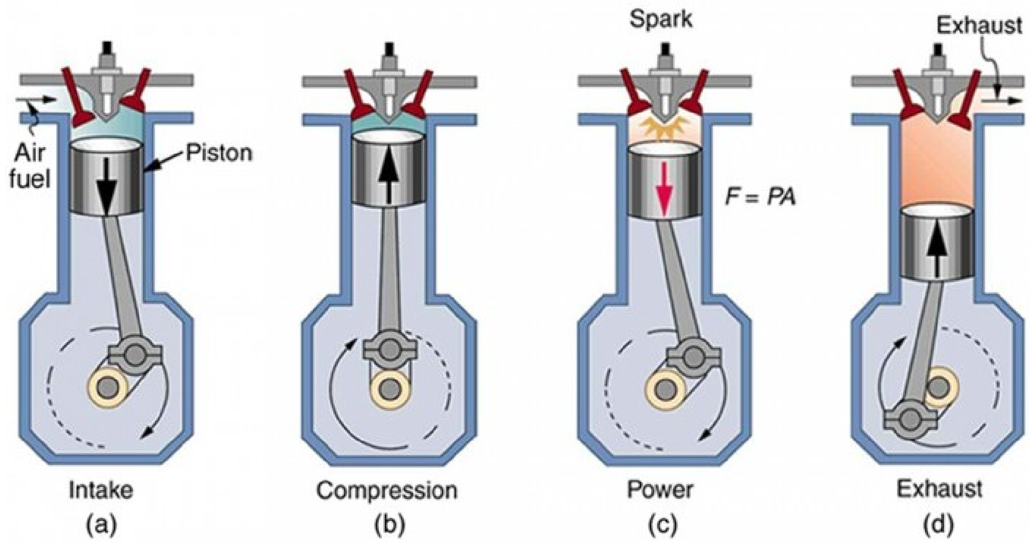
1000 Wh = 1000 Cal



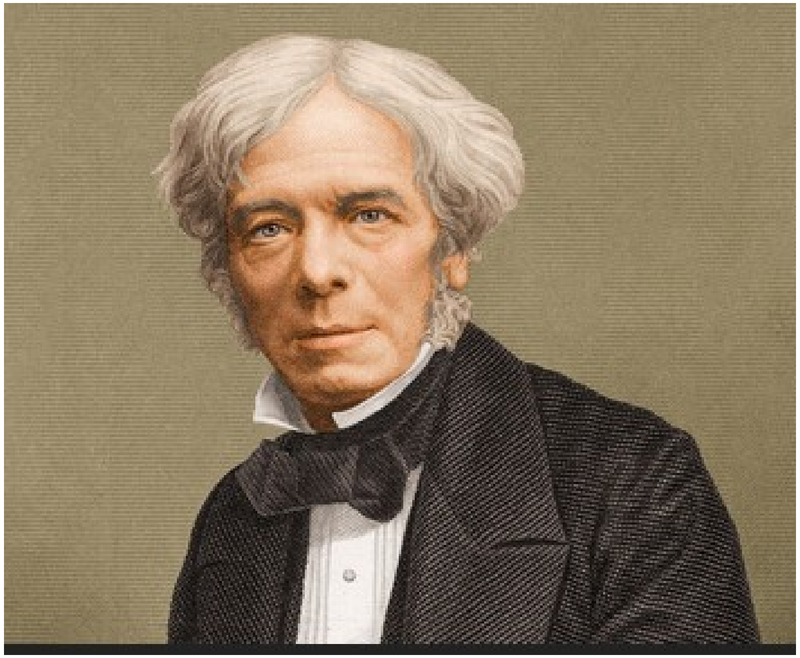

12 micron

Thermodynamics → Science of the study of the flow of heat



**Michael Farady**



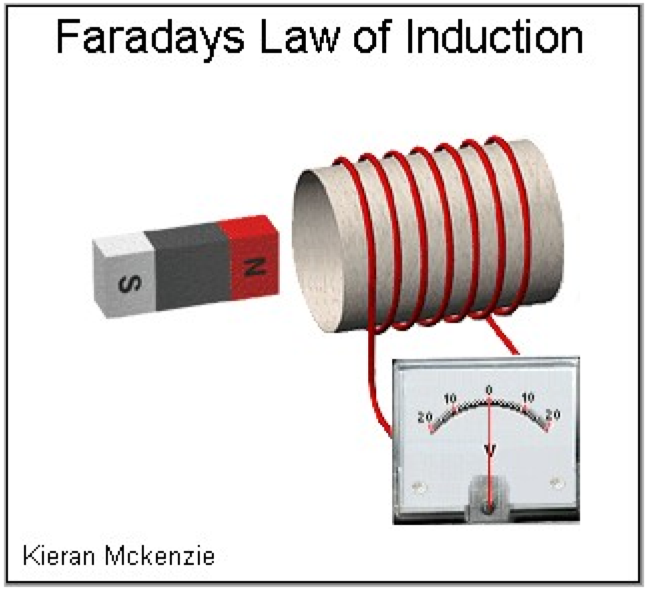


**Figure 0.8** A magnet levitates above a superconductor, demonstrating the Meissner effect. Photo courtesy of Brookhaven National Laboratory

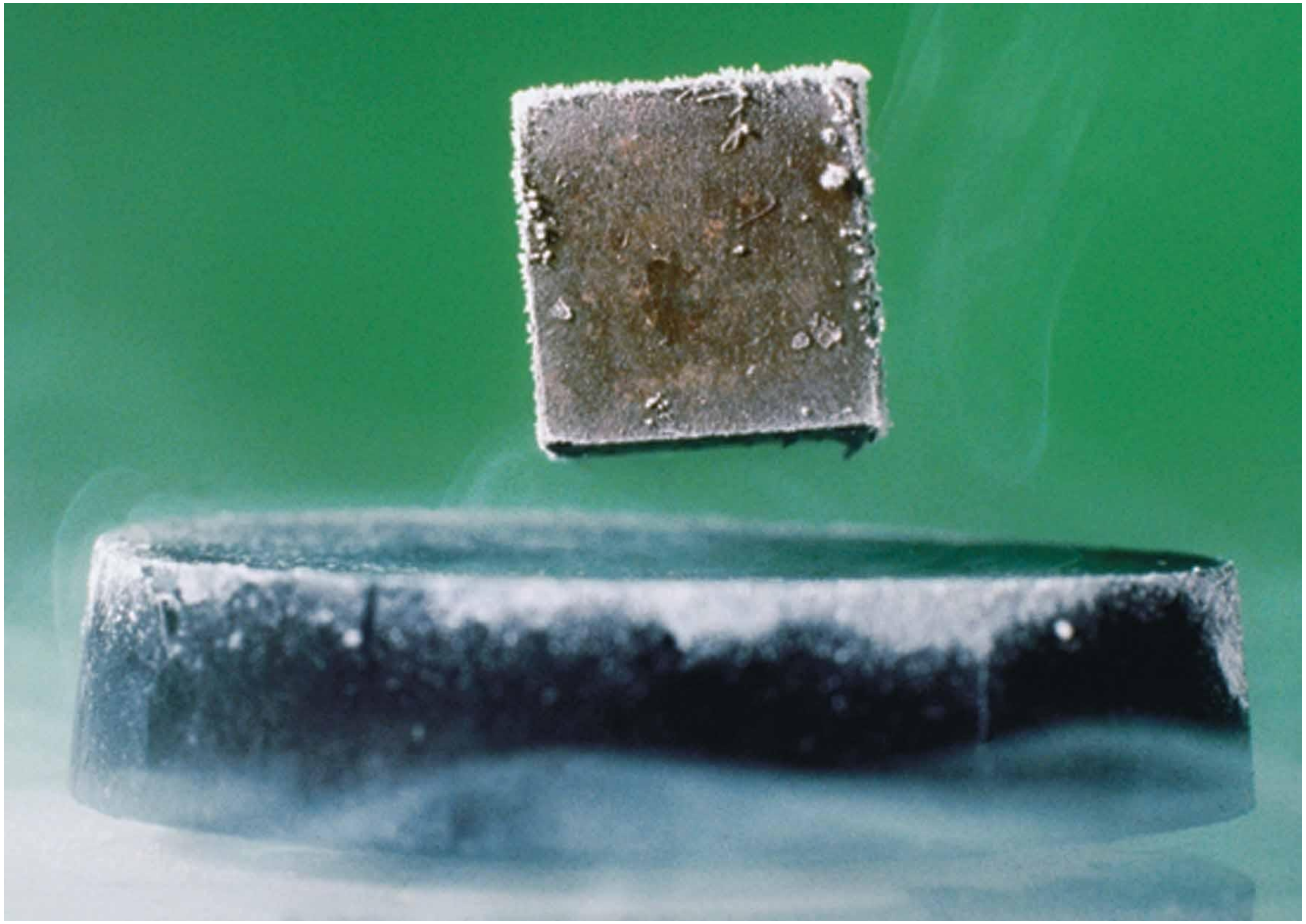


**Figure 0.11** Maglev train technology has high-speed trains competing with airplane service.

**Figure 0.7** The New Clark Bridge, a cable-stayed bridge in Alton, Illinois

Properties of Waves →
resonance or destructive interference



Physics of the very small → transistors
Quantum Physics

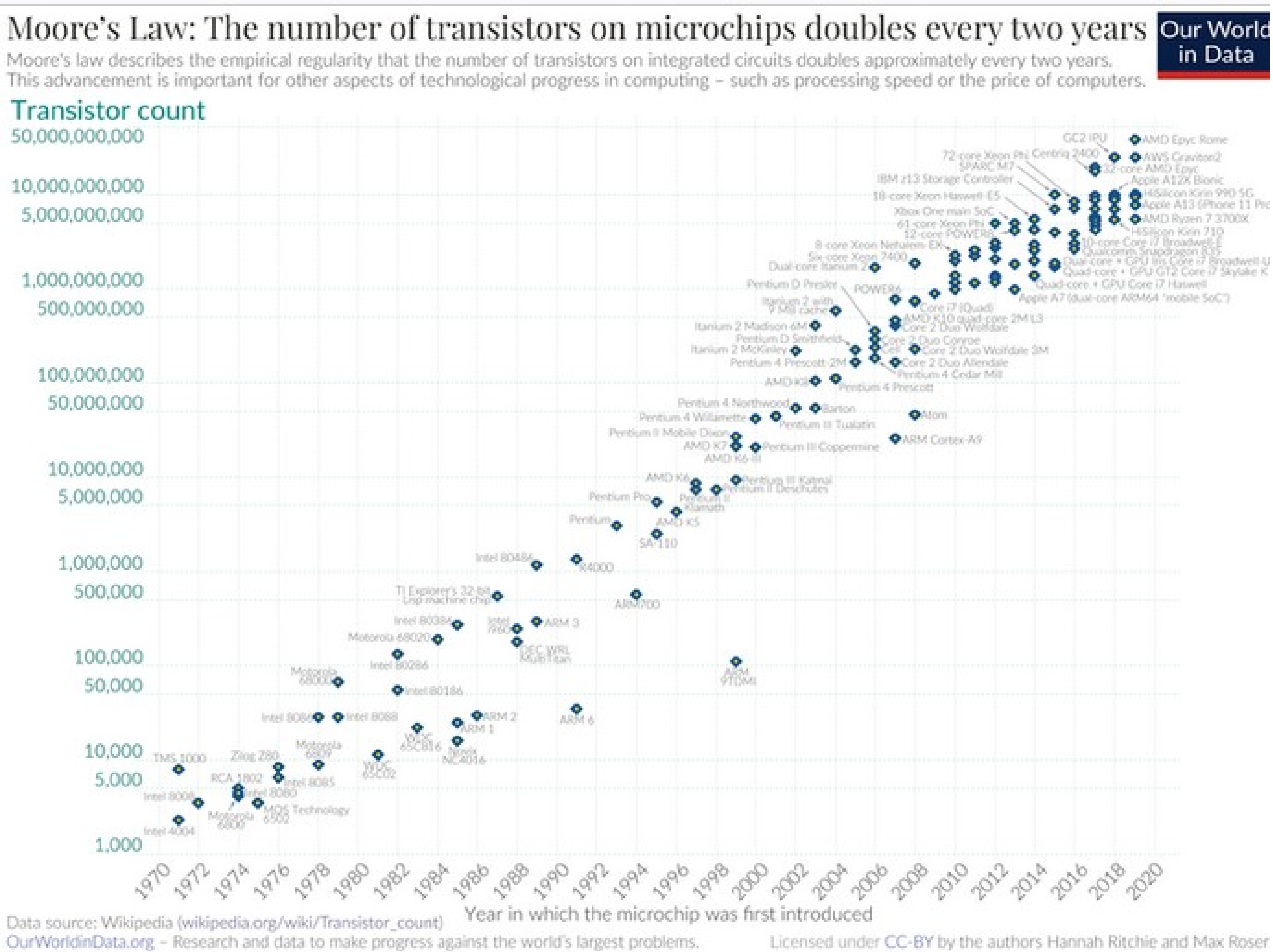
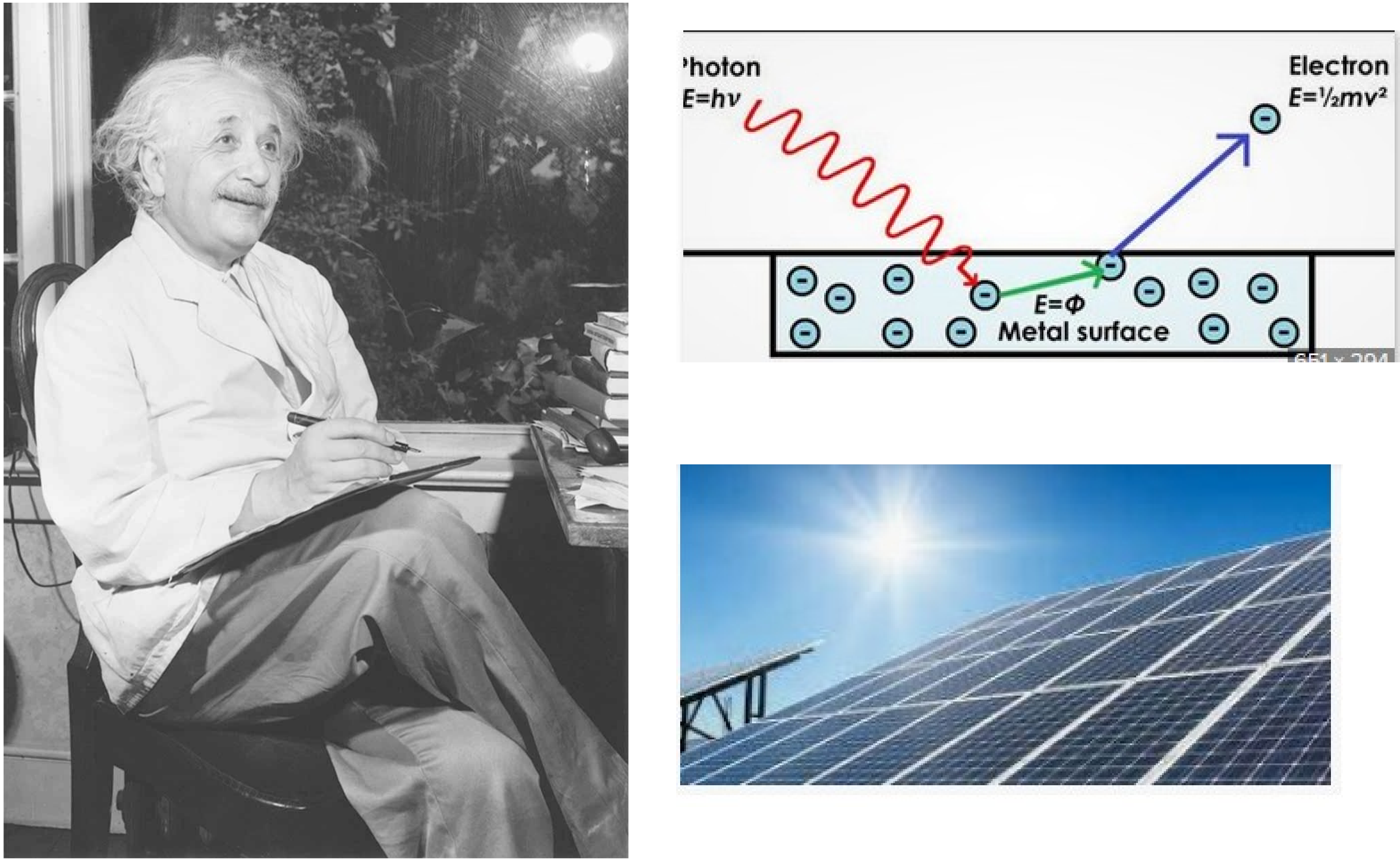

Moore's Law: The number of transistors on microchips doubles every two years
Our World in Data
Moore's law describes the empirical regularity that the number of transistors on integrated circuits doubles approximately every two years.
This advancement is important for other aspects of technological progress in computing – such as processing speed or the price of computers.
Transistor count
50,000,000,000
10,000,000,000
5,000,000,000
1,000,000,000
500,000,000
100,000,000
50,000,000
10,000,000
5,000,000
1,000,000
500,000
100,000
50,000
10,000
5,000
1,000
1970 1972 1974 1976 1978 1980 1982 1984 1986 1988 1990 1992 1994 1996 1998 2000 2002 2004 2006 2008 2010 2012 2014 2016 2018 2020
Year in which the microchip was first introduced
Data source: Wikipedia (wikipedia.org/wiki/Transistor_count)
OurWorldinData.org – Research and data to make progress against the world's largest problems.
Licensed under CC-BY by the authors Hannah Ritchie and Max Roser

**Figure 0.1** Physics is involved in all aspects of cellular phone technology. It controls everything from the electrical circuits in the phone to the transmission of radio waves between phones.

**Figure 0.3** Albert Einstein (1879–1955) is often considered one of the most influential scientists of the twentieth century. His work on relativity, as made famous through the equation $E = mc^2$, and the photoelectric effect changed the way the world viewed physics. Photo courtesy of the National Archives and Records Administration



# BEWARE of pseudo Sciences !

PHOTO: ÉTIENNE KLEIN

A photo tweeted by a famous French physicist supposedly of Proxima Centauri by the James Webb Space Telescope was actually a slice of chorizo.

@EtienneKlein · **Follow**

Photo (VRAIE cette fois...) de la galaxie de la Roue du Chariot et de ses galaxies compagnes, prise par le JWST. Située à 500 millions d'années-lumière, elle fut sans doute spirale dans son passé, mais a pris cette étrange allure à la suite d'un furieux carambolage galactique.